

第 11 章 无穷级数

主要内容

$$\sum_{n=1}^{\infty} u_n \quad (u_n \text{ 为常数})$$

定义	$\sum_{n=1}^{\infty} u_n$ 的收敛, 发散, 绝对收敛与条件收敛	
收敛条件	必要条件	$u_n \rightarrow 0 (n \rightarrow \infty)$
	充要条件	$u_n \geq 0, \{S_n\}$ 有界
重要例子	$\sum_{n=0}^{+\infty} ar^n (a \neq 0)$ $\begin{cases} \text{收敛} & \text{当 } r < 1 \\ \text{发散} & \text{当 } r \geq 1 \end{cases} \quad s = \frac{a}{1-r}$	$\sum_{n=1}^{\infty} \frac{1}{n^p} \begin{cases} \text{收敛} & p > 1 \\ \text{发散} & p \leq 1 \end{cases}$

常数项级数
审敛法

正项级数

任意项级数

1. 若 $S_n \rightarrow S$, 则级数收敛;
2. 当 $n \rightarrow \infty, u_n \rightarrow 0$, 则级数发散;
3. 按基本性质;

4. 充要条件
5. 比较法
6. 比值法
7. 根值法

4. 绝对收敛
5. 交错级数
(莱布尼茨定理)

判 $\sum u_n$ 敛散性的步骤:

$$1. \lim_{n \rightarrow \infty} u_n \begin{cases} \neq 0 & \text{发散} \\ = 0 & \text{进一步讨论} \end{cases}$$

2. 若为同号级数采用 :

(1) 比较法 (比较对象选择: $\sum_{n=1}^{\infty} \frac{1}{n^p}, \sum_{n=0}^{\infty} ar^n$)

(2) 比值法 ($\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n}$)

(3) 根值法 ($\lim_{n \rightarrow \infty} \sqrt[n]{u_n}$)

3. 若为任意项级数

1⁰ 考虑 $\sum |u|$ $\begin{cases} \text{收敛} \\ \text{发散} \end{cases}$ 进一步讨论

2⁰ 考虑 $\sum u_n$ 采用：

(1) $\lim_{n \rightarrow \infty} S_n$; (2) 性质;

(3) 莱布尼茨定理 (仅适用交错级数).

举例

例1 证明: 对于 $\{u_n\}$, 已知 $\lim_{n \rightarrow \infty} nu_n = 0$

则 $\sum_{n=1}^{\infty} (n+1)(u_{n+1} - u_n)$ 收敛 $\Leftrightarrow \sum_{n=1}^{\infty} u_n$ 收敛.

证明

$$\begin{aligned}\sigma_n &= \sum_{k=1}^n (k+1)(u_{k+1} - u_k) \\&= 2(u_2 - u_1) + 3(u_3 - u_2) + \cdots + (n+1)(u_{n+1} - u_n) \\&= -2u_1 - u_2 - u_3 - \cdots - u_n + (n+1)u_{n+1} \\&= -\sum_{k=1}^n u_k - u_1 + (n+1)u_{n+1} \\&= -S_n - u_1 + (n+1)u_{n+1}\end{aligned}$$

$\therefore \lim_{n \rightarrow \infty} \sigma_n$ 与 $\lim_{n \rightarrow \infty} S_n$ 极限同时 $\exists$ 或不 $\exists$

例2 $\sum_{n=2}^{+\infty} (\sqrt{n+1} - \sqrt{n})^p \frac{1}{n-1}$ 敛散性

解 $u_n = \frac{1}{(\sqrt{n+1} + \sqrt{n})^p} \frac{1}{n-1}$

取 $v_n = \frac{1}{n^{\frac{p}{2}+1}}$

$$\frac{u_n}{v_n} = \frac{n^{\frac{p}{2}+1}}{(\sqrt{n+1} + \sqrt{n})^p} \frac{1}{n-1} \rightarrow \frac{1}{2^p} (n \rightarrow \infty)$$

$\therefore p > 0$ 收, $p \leq 0$ 散.

例3 设正项级数 $\sum_{n=1}^{\infty} u_n$ 与 $\sum_{n=1}^{\infty} v_n$ 均收敛, 试证:

$$\sum_{n=1}^{\infty} \sqrt{u_n v_n} \text{ 收敛, } \sum_{n=1}^{\infty} \frac{\sqrt{u_n}}{n} \text{ 也收敛。}$$

证明 $\because \sum_{n=1}^{\infty} u_n$ 与 $\sum_{n=1}^{\infty} v_n$ 均收, $\therefore \sum_{n=1}^{\infty} (u_n + v_n)$ 收敛

$$\text{又 } \because u_n + v_n \geq 2\sqrt{u_n v_n}, \therefore \sum_{n=1}^{\infty} \sqrt{u_n v_n} \text{ 收敛}$$

$$\text{令 } v_n = \frac{1}{n^2}, \because \sum_{n=1}^{\infty} v_n = \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ 收}$$

$$\therefore \text{由上述推导可知 } \sum_{n=1}^{\infty} \frac{\sqrt{u_n}}{n} \text{ 收敛。}$$

例4

$$\sum_{n=1}^{\infty} \frac{n! 2^n \sin \frac{n\pi}{5}}{n^n}$$

解

$$\because \left| \frac{n! 2^n \sin \frac{n\pi}{5}}{n^n} \right| < \frac{n! 2^n}{n^n}$$

$$\text{又 } \frac{(n+1)! 2^{n+1}}{(n+1)^{n+1}} / \frac{n! 2^n}{n^n} = 2 \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{2}{e} < 1 (n \rightarrow \infty)$$

$$\therefore \sum_{n=1}^{\infty} \frac{n! 2^n}{n^n} \text{收,}$$

$$\text{故 } \sum_{n=1}^{\infty} \frac{n! 2^n \sin \frac{n\pi}{5}}{n^n} \text{绝对收敛.}$$

例5 $\sum_{n=2}^{\infty} (-1)^{n-1} \frac{\ln n}{n^p} \quad (p > 0)$

解

$$\text{令 } u_n = \frac{\ln n}{n^p}, \sum_{n=1}^{\infty} \frac{\ln n}{n^p}$$

$$p \leq 1 \quad \sum_{n=2}^{\infty} \frac{\ln n}{n^p} \quad \because \frac{\ln n}{n^p} > \frac{1}{n} (n \geq 3)$$

$$\therefore \sum_{n=1}^{\infty} \frac{\ln n}{n^p} \text{ 发散.}$$

$$0 < p \leq 1 \quad \sum_{n=2}^{\infty} (-1)^{n-1} \frac{\ln n}{n^p}$$

$$\because \left(\frac{\ln x}{x^p}\right)' = \frac{x^{p-1}(1 - p \ln x)}{x^{2p}} < 0, \quad (x > e^{\frac{1}{p}})$$

$$\text{又 } \because \lim_{n \rightarrow \infty} \frac{\ln n}{n^p} = 0 \therefore \sum_{n=2}^{\infty} (-1)^{n-1} \frac{\ln n}{n^p} \text{ 条件收敛.}$$

$$p > 1, \text{ 取 } \sigma \quad \partial \quad p > \sigma > 1, \text{ 令 } v_n = \frac{1}{n^\sigma}$$

$$\frac{u_n}{v_n} = \frac{\ln n}{n^p} \bigg/ \frac{1}{n^\sigma} = \frac{\ln n}{n^{p-\sigma}} \rightarrow 0 (n < +\infty)$$

$$\therefore \sum \frac{1}{n^\sigma} \text{ 收, } \sum \frac{\ln n}{n^p} \text{ 收, 原级数绝对收敛.}$$

例6

讨论 $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n+1} \frac{x^n}{n} + \cdots$
的收敛性。

$|x| < 1$, 原级数绝对收敛

$|x| > 1$, 原级数发散

$x = 1$, 原级数条件收敛

$x = -1$, 原级数发散

课堂练习

1. 判定下列级数的敛散性

$$(1) \cos \frac{\pi}{2} + \cos \frac{\pi}{4} + \cos \frac{\pi}{8} + \cdots + \cos \frac{\pi}{2^n} + \cdots$$

散($\because \lim \cos \frac{\pi}{2^n} = 1 \neq 0$)

$$(2) \frac{1}{2} - \frac{2}{3} + \frac{3}{4} - \frac{4}{5} + \cdots + (-1)^{n-1} \frac{n}{n+1} + \cdots$$

散($\because \lim \frac{n}{n+1} = 1 \neq 0$)

$$(3) \quad \sum_{n=1}^{\infty} \frac{\sin \frac{\pi}{n}}{n} \quad \text{收}(\because \frac{\sin \frac{\pi}{n}}{n} \sim \frac{\pi}{n^2})$$

$$(4) \quad \sum_{n=1}^{\infty} \frac{4^n}{5^n \pm 3^n} \quad \text{收}(\because \frac{4^n}{5^n \pm 3^n} \sim (\frac{4}{5})^n \text{ 或 } \lim \frac{u_{n+1}}{u_n} = \frac{4}{5})$$

$$(5) \quad \sum_{n=2}^{\infty} (\sqrt{n+1} - \sqrt{n}) \frac{1}{n-1}$$

收($\because u_n = \frac{1}{(\sqrt{n+1} + \sqrt{n})(n-1)}$ 与 $\frac{1}{n^{3/2}}$ 比较)

$$(6) \quad \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \ln \frac{n+1}{n-1} \quad \text{收}(\because \frac{1}{\sqrt{n}} \ln(1 + \frac{2}{n-1}) \sim \frac{1}{\sqrt{n}} \frac{2}{n-1})$$

$$(7) \quad \sum_{n=1}^{\infty} \frac{(n!)^2}{2^{n^2}} \quad \text{收} (\because \lim \frac{u_{n+1}}{u_n} = 0 < 1)$$

$$(8) \quad \sum_{n=1}^{\infty} \frac{n^2}{(2 + \frac{1}{n})^n} \quad \text{收(柯西)} \quad \lim \sqrt[n]{u_n} = \frac{1}{2}$$

$$(9) \quad \sum_{n=2}^{\infty} \frac{(-1)^n}{\ln^2 n} \quad \text{收} (\because \text{交错级数, 满足莱布尼茨定理的2个条件})$$

$$(10) \quad \sum_{n=1}^{+\infty} \frac{n \cos \frac{n\pi}{5}}{2^n}$$

$$(|a_n| \leq \frac{n}{2^n} = b_n \quad \lim \frac{b_{n+1}}{b_n} = \frac{1}{2} < 1 \therefore \text{收.})$$

(11). 设 a 为常数, $\sum_{n=1}^{\infty} \left(\frac{\sin na}{n^2} - \frac{1}{\sqrt{n}} \right)$

($\because \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ 散, $\sum_{n=1}^{\infty} \frac{\sin na}{n^2}$ 收 $\therefore$ 原级数发散)

$$(12) \frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}+1} + \cdots + \frac{1}{\sqrt{n}-1} - \frac{1}{\sqrt{n}+1} + \cdots$$

$$\text{散} \left(\sum_{n=2}^{\infty} \left(\frac{1}{\sqrt{n}-1} - \frac{1}{\sqrt{n}+1} \right) \right) = \sum_{n=2}^{\infty} \frac{2}{n-1} = \sum_{n=1}^{\infty} \frac{2}{n}$$

二、(1) $\sum_{n=1}^{\infty} \ln(1 + \frac{1}{n^{\sigma}})$ (σ 为常数) 的敛散性

(2) $\sum_{n=1}^{\infty} \frac{1}{1 + a^n} \quad (a > 0)$

解 (1) 法1: $\sigma \leq 0 \Rightarrow$ 级数发散

$$\sigma > 0, \lim_{n \rightarrow \infty} \frac{u_n}{\frac{1}{n^{\sigma}}} = \lim_{x \rightarrow +\infty} \frac{\ln(1 + \frac{1}{n^{\sigma}})}{\frac{1}{n^{\sigma}}} = 1$$

$\therefore \sum u_n$ 当 $0 < \sigma \leq 1$ 发散, $\sigma > 1$ 收敛

$$\text{法2, } \because \frac{x}{1+x} < \ln(1+x) < x,$$

$$\therefore \frac{1}{1+n^\sigma} < \ln\left(1+\frac{1}{n^\sigma}\right) < \frac{1}{n^\sigma}$$

解(2) $a \leq 1, \lim_{n \rightarrow \infty} \frac{1}{1+a^n} \neq 0 \quad \sum_{n=1}^{\infty} \frac{1}{1+a^n} \text{ 发散}$

$$a > 1, \frac{1}{1+a^n} < \frac{1}{a^n},$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{a^n} \text{ 收敛, } \therefore \sum_{n=1}^{\infty} \frac{1}{1+a^n} \text{ 收敛。}$$

主要内容

幂级数及其收敛区间

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n \quad \left(\sum_{n=0}^{\infty} a_n x^n \right) \quad (a_n \text{ 为幂级数的系数})$$

定义

幂级数的收敛域称为幂级数的**收敛区间**.
收敛区间的半径称为**收敛半径**.

收敛半径求法

$$\text{若 } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho$$

(或 $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \rho$)

$$R = \begin{cases} \frac{1}{\rho} & \rho \neq 0 \\ +\infty & \rho = 0 \\ 0 & \rho = +\infty \end{cases}$$

注意

收敛区间的端点需要单独考虑。

二、求和函数 $S(x)$

求 $\sum_{n=0}^{\infty} a_n x^n$ 的和函数的基本思想是：

1) 记住几何级数的和： $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \quad (|r| < 1)$

利用幂级数的性质（逐项求导或逐项积分）
将所给幂级数化为几何级数求和。

2) 记住 $\sin x$ 、 $\ln(1+x)$ 、 e^x 等的展开式，利用幂级数的性质（逐项求导或逐项积分）将所给幂级数化为这些展开式求和。

三、函数展开成幂级数

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots \\ + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \cdots$$

特别：当 $x_0 = 0$ 时，

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}x^n \\ + \cdots$$



- (1) 弄清函数在哪一点展开； (2) 用间接法
(3) 记住以下函数的展开式：

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots + \frac{1}{n!}x^n + \dots$$

$$x \in (-\infty, +\infty) \text{ --- (1)}$$

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots + \frac{(-1)^{n-1}}{(2n-1)!}x^{2n-1} + \dots$$

$$x \in (-\infty, +\infty) \text{ --- (2)}$$

$$\cos x = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \dots + \frac{(-1)^{k-1}x^{(2k-2)}}{(2k-2)!} + \frac{(-1)^kx^{2k}}{(2k)!} + \dots$$

$$x \in (-\infty, +\infty) \text{ --- (3)}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^n}{n} \quad (-1 < x \leq 1) \quad (4)$$

$$\begin{aligned}
 (1+x)^m &= 1 + mx + \frac{m(m-1)}{2!}x^2 \\
 &\quad + \dots + \frac{m(m-1)(m-2)\cdots(m-n+1)}{n!}x^n + \dots
 \end{aligned}$$

$$(-1 < x < 1) \text{ --- (5)}$$

举

例

例1 求 $2x - \frac{4}{3!}x^3 + \frac{6}{5!}x^5 - \frac{8}{7!}x^7 + \cdots$ 的收敛域与和函数。

解 令 $s(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2nx^{2n-1}}{(2n-1)!},$

两边逐项求积分得

$$\int_0^x s(x) dx = \int_0^x \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2nx^{2n-1}}{(2n-1)!} dx$$

$$\begin{aligned}
 \text{即 } \int_0^x s(x) dx &= \int_0^x \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2nx^{2n-1}}{(2n-1)!} dx \\
 &= \sum_{n=1}^{\infty} (-1)^{n-1} \int_0^x \frac{2nx^{2n-1}}{(2n-1)!} dx \\
 &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{(2n-1)!} = x \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} \\
 &= x \sin x \quad x \in (-\infty, +\infty)
 \end{aligned}$$

$$\text{故 } s(x) = (x \sin x)'$$

$$= \sin x + x \cos x \quad x \in (-\infty, +\infty)$$

例2 (1)将 $f(x) = \frac{1}{4} \ln \frac{1+x}{1-x} + \frac{1}{2} \arctan x - x$
展开为的幂级数.

(2)将 $\ln \frac{x}{1+x}$ 展开为 $(x-1)$ 的幂级数。

解(1) $f'(x) = \frac{1}{4} \left(\ln \frac{1+x}{1-x} \right)' + \frac{1}{2} (\arctan x)' - 1$

$$= \frac{1}{4} \left[\frac{1}{1+x} + \frac{1}{1-x} \right] + \frac{1}{2} \frac{1}{1+x^2} - 1 = \frac{x^4}{1-x^4}$$

$$= \sum_{n=1}^{\infty} x^{4n} \quad -1 < x < 1$$

对上式两边积分得

$$f(x) - f(0) = \int_0^x f'(x) dx$$

$$= \int_0^x \sum_{n=1}^{\infty} x^{4n} dx = \sum_{n=1}^{\infty} \frac{x^{4n+1}}{4n+1} \quad (-1 < x < 1)$$

$$\because f(0) = 0 \quad \therefore f(x) = \sum_{n=1}^{\infty} \frac{x^{4n+1}}{4n+1} \quad (-1 < x < 1)$$

$$\begin{aligned}\text{解(2)} \quad \ln \frac{x}{1+x} &= \ln x - \ln(1+x) \\ &= \ln[1+(x-1)] - \ln[2+(x-1)]\end{aligned}$$

$$\therefore \ln[1+(x-1)] = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n}$$

$$-1 < x-1 \leq 1 \quad \text{即}, x \in (0, 2]$$

$$\text{而} \quad \ln[2+(x-1)] = \ln 2 + \ln\left[1 + \frac{x-1}{2}\right]$$

$$= \ln 2 + \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n 2^n}$$

$$-1 < \frac{x-1}{2} \leq 1 \quad \text{即}, x \in (-1, 3]$$

$$\therefore \ln \frac{x}{1+x} = -\ln 2 + \sum_{n=1}^{\infty} (-1)^{n-1} \left(1 - \frac{1}{2^n}\right) \frac{(x-1)^n}{n} \quad x \in (0, 2]$$

例3 将级数 $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2^{n-1}} \cdot \frac{x^{2n-1}}{(2n-1)!}$ 的和函数展开成 $(x-1)$ 的幂级数.

解 分析 $\because \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!}$ 是 $\sin x$ 的展开式,
设法用已知展开式来解.

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2^{n-1}} \cdot \frac{x^{2n-1}}{(2n-1)!} &= \sqrt{2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)!} \left(\frac{x}{\sqrt{2}}\right)^{2n-1} \\ &= \sqrt{2} \sin \frac{x}{\sqrt{2}} = \sqrt{2} \sin \frac{x-1+1}{\sqrt{2}} \end{aligned}$$

$$\begin{aligned}
&= \sqrt{2} \sin \frac{1}{\sqrt{2}} \cos \frac{x-1}{\sqrt{2}} + \sqrt{2} \cos \frac{1}{\sqrt{2}} \sin \frac{x-1}{\sqrt{2}} \\
&= \sqrt{2} \sin \frac{1}{\sqrt{2}} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{x-1}{\sqrt{2}} \right)^{2n} \\
&\quad + \sqrt{2} \cos \frac{1}{\sqrt{2}} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \left(\frac{x-1}{\sqrt{2}} \right)^{2n+1} \\
&= \sqrt{2} \sin \frac{1}{\sqrt{2}} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n \cdot (2n)!} (x-1)^{2n} \\
&\quad + \cos \frac{1}{\sqrt{2}} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^n (2n+1)!} (x-1)^{2n+1} \quad (-\infty, +\infty)
\end{aligned}$$

注意：利用 $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$ ，求一些数项级数的和。

例4 求 $\sum_{n=1}^{\infty} \frac{n^2}{n!}$ 的和。

解 令 $s(x) = \sum_{n=1}^{\infty} \frac{n^2}{n!} x^n = \sum_{n=2}^{\infty} \frac{1}{(n-2)!} x^n + x \sum_{n=1}^{\infty} \frac{1}{(n-1)!} x^{n-1}$
 $(-\infty, +\infty)$

$$\because s(x) = \sum_{n=1}^{\infty} \frac{n}{(n-1)!} x^n = e^x x^2 + x e^x,$$

$$= \sum_{n=2}^{\infty} \frac{(n-1)}{(n-1)!} x^n + \sum_{n=1}^{\infty} \frac{1}{(n-1)!} x^n \quad \sum_{n=1}^{\infty} \frac{n^2}{n!} = s(1) = 2e.$$

例5 求 $\sum_{n=1}^{\infty} \frac{n^2}{n!2^n}$ 的和.

解 令 $s(x) = \sum_{n=1}^{\infty} \frac{n^2}{n!} x^n$, $(-\infty, +\infty)$

$$\because s(x) = \sum_{n=1}^{\infty} \frac{n(n-1) + n}{n!} x^n = \sum_{n=1}^{\infty} \frac{n(n-1)}{n!} x^n + \sum_{n=1}^{\infty} \frac{1}{(n-1)!} x^n$$

$$= x^2 \left(\sum_{n=1}^{\infty} \frac{x^n}{n!} \right)'' + x \sum_{n=0}^{\infty} \frac{x^n}{n!} = x^2 (e^x - 1)'' + x e^x$$

$$= e^x (x+1)x,$$

$$\therefore \sum_{n=1}^{\infty} \frac{n^2}{n!2^n} = s\left(\frac{1}{2}\right) = e^{\frac{1}{2}} \left(\frac{1}{2} + 1\right) \frac{1}{2} = \frac{3}{4} \sqrt{e}.$$

例6 求幂级数 $\sum_{n=1}^{\infty} \frac{n^2 + 1}{2^n n!} x^n$ 的和函数.

提示:

$$\sum_{n=0}^{\infty} \frac{n^2 + 1}{2^n n!} x^n = \sum_{n=0}^{\infty} \frac{n^2}{n!} \left(\frac{x}{2}\right)^n + \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{x}{2}\right)^n$$

例7 将 $\sin x$ 展成 $(x - \frac{\pi}{4})$ 的幂级数

解 $\sin x = \sin(\frac{\pi}{4} + (x - \frac{\pi}{4}))$

$$= \sin \frac{\pi}{4} \cos(x - \frac{\pi}{4}) + \cos \frac{\pi}{4} \sin(x - \frac{\pi}{4})$$

$$= \frac{1}{\sqrt{2}} [\cos(x - \frac{\pi}{4}) + \sin(x - \frac{\pi}{4})]$$

$$\therefore \sin x = \frac{1}{\sqrt{2}} [1 + (x - \frac{\pi}{4}) - \frac{(x - \frac{\pi}{4})^2}{2!} - \frac{(x - \frac{\pi}{4})^3}{3!} + \frac{(x - \frac{\pi}{4})^4}{4!} + \frac{(x - \frac{\pi}{4})^5}{5!} + \cdots]$$

$$-\infty < x < +\infty$$

EX 1. 将 $y = \cosh x$ 展成 Maclaurin 级数。

$$\frac{1}{2}(e^x + e^{-x}) = \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$(-\infty, +\infty)$

2. 设 $y(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$, 则 $f^{(k)}(0) = \underline{\hspace{2cm}}$

$$= \begin{cases} \frac{(-1)^n}{(2n+1)!} & k = 2n \\ 0 & k = 2n+1 \end{cases} \quad n = 0, 1, \dots$$

课堂练习

1. 求级数 $\sum_{n=0}^{\infty} (n+1)(x-1)^n$ 收敛域及和函数.

解 $\because \sum_{n=0}^{\infty} (n+1)(x-1)^n$ 的收敛半径为 $R=1$,

收敛域为 $-1 < x-1 < 1$, 即 $0 < x < 2$,

设此级数的和函数为 $s(x)$, 则有

$$s(x) = \sum_{n=0}^{\infty} (n+1)(x-1)^n.$$

两边逐项积分

$$\begin{aligned}
 \int_1^x s(x) dx &= \sum_{n=0}^{\infty} \int_1^x (n+1)(x-1)^n dx \\
 &= \sum_{n=0}^{\infty} (x-1)^{n+1} \Big|_1^x = \sum_{n=0}^{\infty} (x-1)^{n+1} \\
 &= \frac{x-1}{1-(x-1)} = \frac{x-1}{2-x},
 \end{aligned}$$

两边再对 x 求导，得

$$s(x) = \left(\frac{x-1}{2-x} \right)' = \frac{1}{(2-x)^2}.$$

2. 求: $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots + \frac{x^{2n}}{(2n)!} + \cdots$ 的收敛域与和函数

解 $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{2n}}{(2n)!} / \frac{x^{2(n-1)}}{(2n-2)!} \right| = 0 < 1 \quad (x \neq 0)$

$$R = +\infty \quad \therefore x \in (-\infty, +\infty)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$$

$$e^x + e^{-x} = \sum_{n=0}^{\infty} \frac{1 + (-1)^n}{n!} x^n = 2 \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\therefore \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = \frac{e^x + e^{-x}}{2} \quad x \in (-\infty, +\infty)$$

3. 将 $f(x) = x \arctan x - \ln \sqrt{1+x^2}$ 展开成麦克劳林级数.

解 $\because \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots,$

$$\therefore \ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \cdots + (-1)^{n-1} \frac{x^{2n}}{n} + \cdots,$$

$(-1 \leq x \leq 1)$

又 $\arctan x = \int_0^x \frac{1}{1+x^2} dx$

$$= \int_0^x [1 - x^2 + x^4 - x^6 + \cdots + (-1)^n x^{2n} + \cdots] dx$$

$$= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots + (-1)^n \frac{x^{2n+1}}{2n+1} + \cdots$$

$$(-1 \leq x \leq 1)$$

故 $x \arctan x - \ln \sqrt{1+x^2}$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+1} - \frac{1}{2} \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n}}{n}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+1} - \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{2n+2}$$

$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+2}}{(2n+1)(2n+2)}. \quad (-1 \leq x \leq 1)$$

4. (1) 将函数 $f(x) = \arctan \frac{1+x}{1-x}$ 展开为麦克劳林级数。

(2) 将函数 $f(x) = \frac{1}{x^2 + 3x + 2}$ 展开为 $(x+4)$ 的泰勒级数。

解 (1) $f'(x) = \left(\arctan \frac{1+x}{1-x} \right)'$

$$= \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n} \quad x \in (-1, 1)$$

$$\begin{aligned} f(x) - f(0) &= \int_0^x \left(\arctan \frac{1+x}{1-x} \right)' dx = \int_0^x \sum_{n=0}^{\infty} (-1)^n x^{2n} dx \\ &= \sum_{n=0}^{\infty} (-1)^n \int_0^x x^{2n} dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad x \in (-1,1) \end{aligned}$$

$$\because f(0) = \arctan 1 = \frac{\pi}{4}$$

$$\begin{aligned} \therefore f(x) = \arctan \frac{1+x}{1-x} &= \frac{\pi}{4} + \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \\ &\quad x \in (-1,1) \end{aligned}$$

$$(2) f(x) = \frac{1}{x^2 + 3x + 2} = \frac{1}{x+1} - \frac{1}{x+2}$$

$$\frac{1}{x+1} = \frac{-1}{3-(x+4)} = -\frac{1}{3} \sum_{n=0}^{\infty} \frac{1}{3^n} (x+4)^n \quad -1 < \frac{x+4}{3} < 1$$

$$\frac{1}{x+2} = \frac{-1}{2-(x+4)} = -\frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} (x+4)^n \quad -1 < \frac{x+4}{2} < 1$$

$$f(x) = \frac{1}{x^2 + 3x + 2} = \sum_{n=0}^{\infty} \left[\frac{1}{2^{n+1}} - \frac{1}{3^{n+1}} \right] (x+4)^n$$

$$x \in (-6, -2)$$

Fourier级数

讨论另一类特殊重要的函数项级数：

由三角函数组成的函数项级数----三角级数

研究：

怎样将一个函数用三角级数表示？

即 如何将一个函数展成三角级数？

4.1 周期函数与三角级数

4.2 三角函数系的正交性与Fourier级数

4.3 周期函数的Fourier展开

4.4 $[0, \eta]$ 上函数的Fourier级数

例1 已知 $f(x)$ 是以 2π 为周期的周期函数, 在 $-\pi \leq x < \pi$ 上 $f(x)$ 的表达式为

$$f(x) = \begin{cases} 0, & -\pi \leq x < 0, \\ x, & 0 \leq x < \pi. \end{cases}$$

设 $s(x)$ 是 $f(x)$ 的傅里叶级数的和函数, 求 $s\left(\frac{\pi}{2}\right), s\left(-\frac{\pi}{2}\right), s(\pi), s\left(\frac{7\pi}{2}\right)$.

解: 由狄利克雷收敛定理知

$$s\left(\frac{\pi}{2}\right) = f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}, \quad s\left(-\frac{\pi}{2}\right) = f\left(-\frac{\pi}{2}\right) = 0,$$

$$s(\pi) = \frac{f(\pi+0) + f(\pi-0)}{2} = \frac{\pi + 0}{2} = \frac{\pi}{2},$$

$$s\left(\frac{7\pi}{2}\right) = s\left(-\frac{\pi}{2}\right) = 0$$

例 2 将函数 $f(x) = x + 1$ ($0 \leq x \leq \pi$) 分别展开成正弦级数和余弦级数.

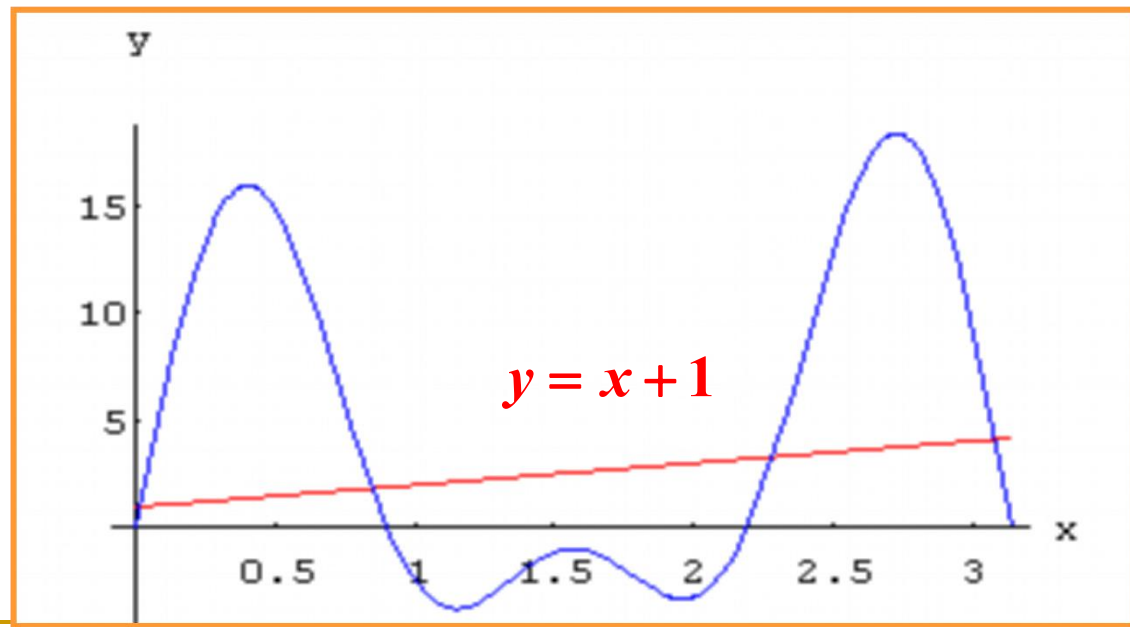
解 (1) 求正弦级数. 对 $f(x)$ 进行奇延拓,

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx = \frac{2}{\pi} \int_0^{\pi} (x + 1) \sin nx dx \\ &= \frac{2}{n\pi} (1 - \pi \cos n\pi - \cos n\pi) \\ &= \begin{cases} \frac{2}{\pi} \cdot \frac{\pi + 2}{n} & \text{当 } n = 1, 3, 5, \dots \\ -\frac{2}{n} & \text{当 } n = 2, 4, 6, \dots \end{cases} \end{aligned}$$

$$x + 1 = \frac{2}{\pi} [(\pi + 2)\sin x - \frac{\pi}{2}\sin 2x + \frac{1}{3}(\pi + 2)\sin 3x - \dots]$$

$(0 < x < \pi)$

$$x + 1 = \frac{2}{\pi} [(\pi + 2)\sin x - \frac{\pi}{2}\sin 2x + \frac{1}{3}(\pi + 2)\sin 3x - \frac{\pi}{4}\sin 4x + \frac{1}{5}(\pi + 2)\sin 5x]$$



(2) 求余弦级数. 对 $f(x)$ 进行偶延拓,

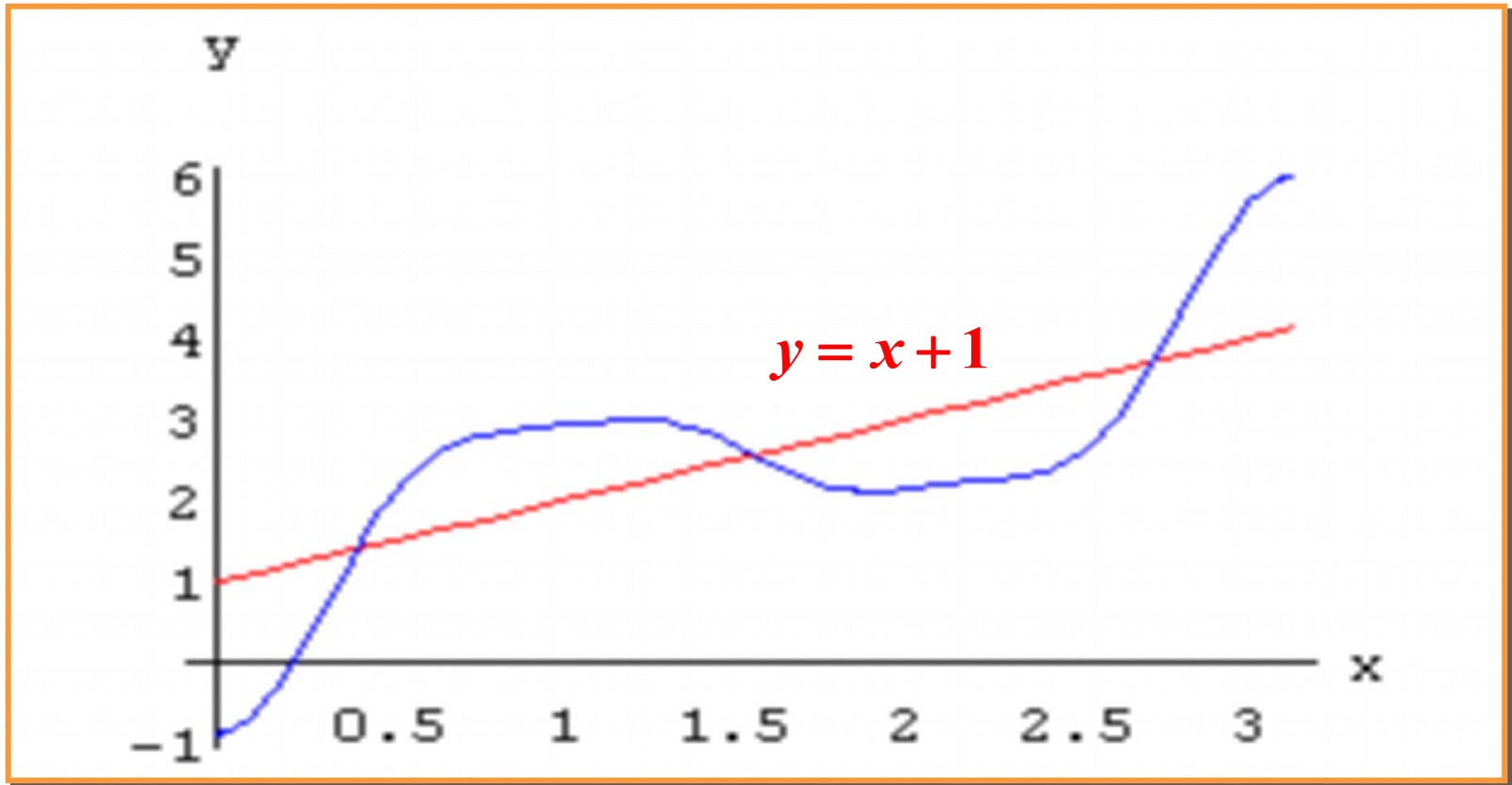
$$a_0 = \frac{2}{\pi} \int_0^{\pi} (x+1) dx = \pi + 2,$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} (x+1) \cos nx dx \\ &= \frac{2}{n^2 \pi} (\cos n\pi - 1) = \begin{cases} 0 & \text{当 } n = 2, 4, 6, \dots \\ -\frac{4}{n^2 \pi} & \text{当 } n = 1, 3, 5, \dots \end{cases} \end{aligned}$$

$$x+1 = \frac{\pi}{2} + 1 - \frac{4}{\pi} \left(\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right)$$

$$\underline{(0 \leq x \leq \pi)}$$

$$x+1 = \frac{\pi}{2} + 1 - \frac{4}{\pi} \left(\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \frac{1}{7^2} \cos 7x \right)$$



例3 证明：当 $0 \leq x \leq \pi$ 时， $\sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = \frac{x^2}{4} - \frac{\pi x}{2} + \frac{\pi^2}{6}$.

解 设 $f(x) = \frac{x^2}{4} - \frac{\pi x}{2}$,

将 $f(x)$ 在 $[0, \pi]$ 上展开成余弦级数：

$$a_0 = \frac{2}{\pi} \int_0^{\pi} \left(\frac{x^2}{4} - \frac{\pi x}{2} \right) dx = \frac{2}{\pi} \left(\frac{\pi^3}{12} - \frac{\pi^3}{4} \right) = -\frac{\pi^3}{3},$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} \left(\frac{x^2}{4} - \frac{\pi x}{2} \right) \cos nx dx$$

$$= \frac{2}{n\pi} \left[\left(\frac{x^2}{4} - \frac{\pi x}{2} \right) \sin nx \right]_0^{\pi} - \int_0^{\pi} \left(\frac{x}{2} - \frac{\pi}{2} \right) \sin nx dx$$

$$= \frac{2}{n^2\pi} \int_0^{\pi} \left(\frac{x}{2} - \frac{\pi}{2} \right) d \cos nx = \frac{2}{n^2\pi} \cdot \frac{\pi}{2} = \frac{1}{n^2}.$$

$$\therefore \frac{x^2}{4} - \frac{\pi x}{2} = -\frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}. \quad (0 \leq x \leq \pi)$$

故
$$\sum_{n=1}^{\infty} \frac{\cos nx}{n^2} = \frac{x^2}{4} - \frac{\pi x}{2} + \frac{\pi^2}{6}.$$

例4 将函数 $f(x) = 2 + |x|$ ($-1 \leq x \leq 1$) 内展开成

以 2 为周期的付氏级数, 并由此求级数 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 的和.

解 $\because f(x) = 2 + |x|$ ($-1 \leq x \leq 1$) 是偶函数,

$$\therefore a_0 = \frac{2}{1} \int_0^1 (2+x) dx = 5,$$

$$a_n = \frac{2}{1} \int_0^1 (2+x) \cos \frac{n\pi x}{1} dx = 2 \int_0^1 x \cos n\pi x dx$$

$$= \frac{2}{n\pi} \int_0^1 x d \sin n\pi x = \frac{2}{n^2 \pi^2} [(-1)^n - 1]$$

$$= \begin{cases} 0, & n = 2k \\ -\frac{4}{n^2 \pi^2}, & n = 2k-1 \end{cases} \quad (k = 1, 2, \dots)$$

$$b_n = 0,$$

$$\begin{aligned} \text{故 } 2 + |x| &= \frac{5}{2} + \sum_{k=1}^{\infty} -\frac{4}{\pi^2 (2k-1)^2} \cos(2k-1)\pi x \\ &= \frac{5}{2} - \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{\cos(2k-1)\pi x}{(2k-1)^2}. \quad (-1 \leq x \leq 1) \end{aligned}$$

$$\text{取 } x = 0, \text{ 由上式得 } 2 = \frac{5}{2} - \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2},$$

$$\therefore \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{\pi^2}{8},$$

$$\text{而 } \sum_{n=1}^{\infty} \frac{1}{n^2} = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} + \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} + \frac{1}{4} \sum_{k=1}^{\infty} \frac{1}{k^2},$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8} \cdot \frac{4}{3} = \frac{\pi^2}{6}.$$

例5. 若函数 $f(x)=x^2$ ($0 \leq x < 1$), 则 $S(x)=\sum_{n=1}^{\infty} b_n \sin n\pi x$ ($-\infty < x < +\infty$), 其中

$$b_n = 2 \int_0^1 f(x) \sin n\pi x dx (n=1,2,\dots), \text{ 则 } S(-\frac{1}{2}) = \quad (B).$$

$$(A) \quad -\frac{1}{2}; \quad (B) \quad -\frac{1}{4}; \quad (C) \quad \frac{1}{4}; \quad (D) \quad \frac{1}{2}.$$

提示：奇延拓展开为正弦级数，并利用狄利克雷定理

奇延拓：

$$\text{则 } F(x) = \begin{cases} f(x) & 0 < x \leq \pi \\ 0 & x = 0 \\ -f(-x) & -\pi < x < 0 \end{cases}$$

例6. 把函数 $f(x) = \begin{cases} 1-x, & 0 < x \leq 2, \\ x-3, & 2 < x < 4 \end{cases}$ 在 $(0, 4)$ 上展开成余弦级数.

解 为把 f 展开成余弦级数, 对 f 作偶式周期延拓, 如图 15-11 所示.

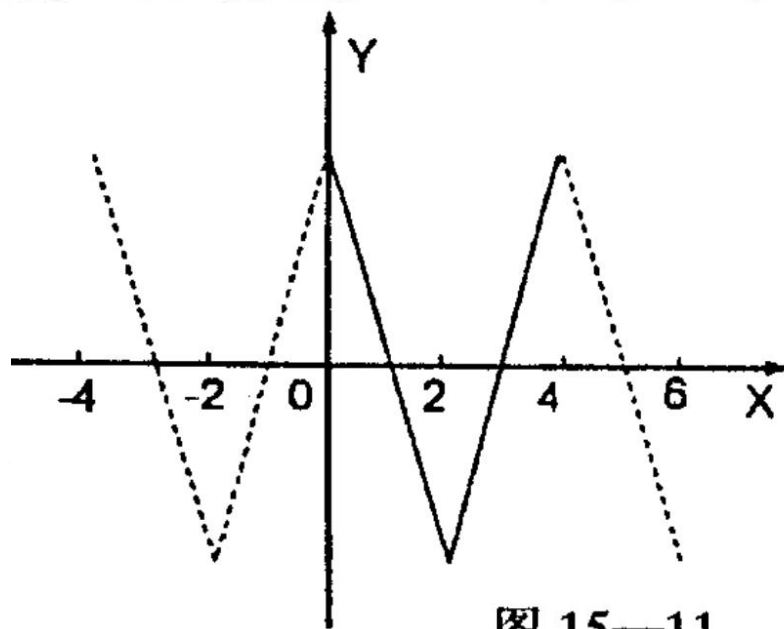


图 15—11

$$a_0 = \frac{2}{4} \int_0^4 f(x) dx = \frac{1}{2} \left[\int_0^2 (1-x) dx + \int_2^4 (x-3) dx \right] = 0$$

$$\begin{aligned}
a_n &= \frac{2}{4} \int_0^4 f(x) \cos \frac{n\pi x}{4} dx \\
&= \frac{1}{2} \left[\int_0^2 (1-x) \cos \frac{n\pi x}{4} dx + \int_2^4 (x-3) \cos \frac{n\pi x}{4} dx \right] \\
&= \frac{1}{2} \left[(1-x) \cdot \frac{4}{n\pi} \sin \frac{n\pi x}{4} - \left(\frac{4}{n\pi} \right)^2 \cos \frac{n\pi x}{4} \right] \Big|_0^2 \\
&\quad + \frac{1}{2} \left[(x-3) \frac{4}{n\pi} \sin \frac{n\pi x}{4} + \left(\frac{4}{n\pi} \right)^2 \cos \frac{n\pi x}{4} \right] \Big|_2^4 \\
&= \left(\frac{4}{n\pi} \right)^2 \left\{ -\cos \frac{n\pi}{2} + \frac{1}{2} [1 + (-1)^n] \right\} \\
&= \begin{cases} 0, & \text{当 } n = 2k-1 \text{ 时,} \\ \frac{4}{k^2 \pi^2} [-(-1)^k + 1], & \text{当 } n = 2k \text{ 时,} \end{cases} \quad k = 1, 2, \dots \\
&= \begin{cases} 0, & \text{当 } n = 2k-1 \text{ 时,} \\ 0, & \text{当 } n = 2k \text{ 且 } k = 2m \text{ 时,} \\ \frac{8}{(2m-1)^2 \pi^2}, & \text{当 } n = 2k \text{ 且 } k = 2m-1 \text{ 时} \end{cases} \quad \text{其中 } m = 1, 2, \dots
\end{aligned}$$

根据收敛定理,在区间(0,4) 上,

$$f(x) = \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos \frac{(2n-1)\pi x}{2}$$

例7. 把函数 $f(x)=(x-1)^2$ 在 $(0, 1)$ 上展开成余弦级数, 并推出 $\pi^2 = 6\left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots\right)$.

解 为把 f 展开成余弦级数, 对 f 作偶式周期延拓, 如图 15-12 所示.

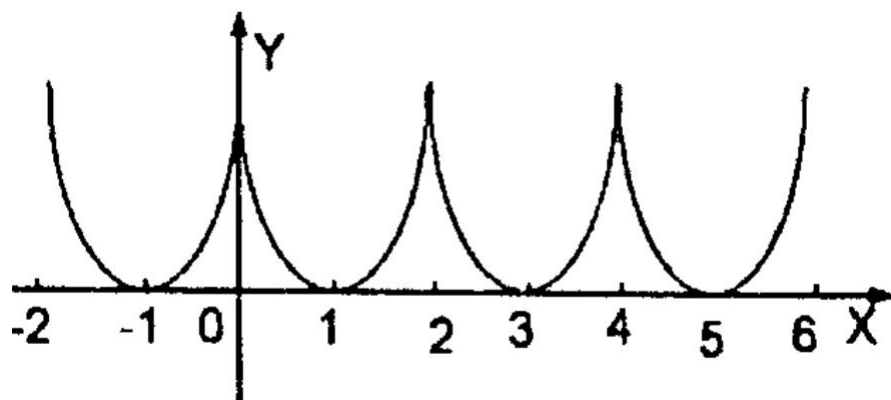


图 15—12

$$\begin{aligned} a_0 &= 2 \int_0^1 (x-1)^2 dx \\ &= \frac{2}{3} \end{aligned}$$

$$a_n = 2 \int_0^1 (x-1)^2 \cos n\pi x dx$$

$$= 2 \left[\frac{1}{n\pi} (x-1)^2 \sin n\pi x \Big|_0^1 - \frac{2}{n\pi} \int_0^1 (x-1) \sin n\pi x dx \right]$$

$$= \frac{4}{n\pi} \cdot \frac{1}{n\pi} \left[(x-1) \cos n\pi x \Big|_0^1 - \int_0^1 \cos n\pi x dx \right] = \frac{4}{n^2 \pi^2}$$

根据收敛定理,在区间 $(0,1)$ 上

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{\cos n\pi x}{n^2}$$

当 $x = 0$ 时,由 f 延拓后连续,可得 $1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2}$

即 $\pi^2 = 6(1 + \frac{1}{2^2} + \frac{1}{3^2} + \cdots)$

例8. 将函数 $f(x) = \cos \frac{x}{2}$ 在 $[0, \pi]$ 上展开正弦级数.

解 为了把 f 展开成正弦级数, 对 f 作奇式周期延拓, 如图 15-10 所示.

$$a_0 = 0, a_n = 0, n = 1, 2, \dots$$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} \cos \frac{x}{2} \sin nx dx \\ &= \frac{2}{\pi} \int_0^{\pi} \frac{1}{2} \left[\sin \left(n + \frac{1}{2} \right) x \right. \\ &\quad \left. + \sin \left(n - \frac{1}{2} \right) x \right] dx \\ &= \frac{1}{\pi} \cdot \frac{2}{2n+1} \end{aligned}$$

$$\begin{aligned} &\left[-\cos \left(n + \frac{1}{2} \right) x \right] \Big|_0^{\pi} + \frac{2}{\pi} \frac{1}{2n-1} \left[-\cos \left(n - \frac{1}{2} \right) x \right] \Big|_0^{\pi} \\ &= \frac{8}{\pi} \cdot \frac{n}{4n^2 - 1} \end{aligned}$$

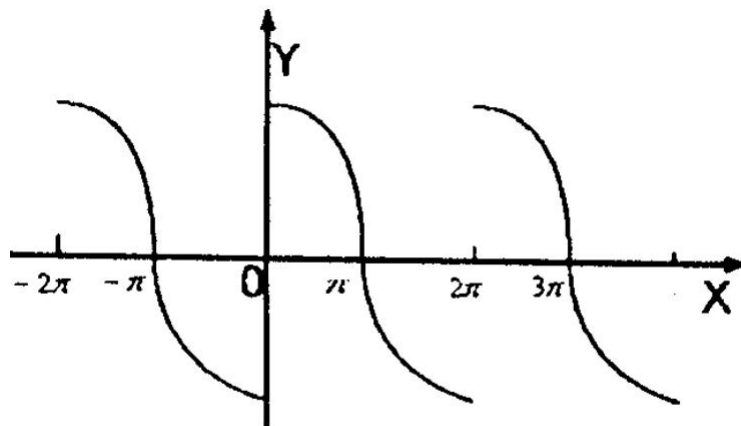


图 15—10

因此,由收敛定理,在区间 $(0, \pi)$ 上:

$$\cos \frac{x}{2} = \frac{8}{\pi} \sum_{n=1}^{\infty} \frac{n}{4n^2 - 1} \sin nx$$

但当 $x = 0, \pi$ 时,右端级数收敛于 0.

注：当函数定义在 $[a,b]$ 上时,且满足狄氏条件

其傅里叶展开方法:

$$f(x), x \in [a, b]$$

$$T = b - a$$

$$\text{令 } z = x - \frac{b+a}{2} \quad \text{即} \quad x = z + \frac{b+a}{2},$$

$$F(z) := f(x) = f\left(z + \frac{b+a}{2}\right), z \in \left[-\frac{b-a}{2}, \frac{b-a}{2}\right]$$

周期延拓

$F(z)$ 在 $\left[-\frac{b-a}{2}, \frac{b-a}{2}\right]$ 上展成傅里叶级数

将 $z = x - \frac{b+a}{2}$ 代入展开式

$f(x)$ 在 $[a, b]$ 上的傅里叶级数

例9 将函数 $f(x) = 10 - x$ ($5 < x < 15$) 展开成傅氏级数.

解 作变量代换 $z = x - 10$,

$$5 < x < 15 \Rightarrow -5 < z < 5,$$

$$f(x) = f(z + 10) = -z = F(z),$$

补充函数 $F(z) = -z$ ($-5 < z < 5$) 的定义,

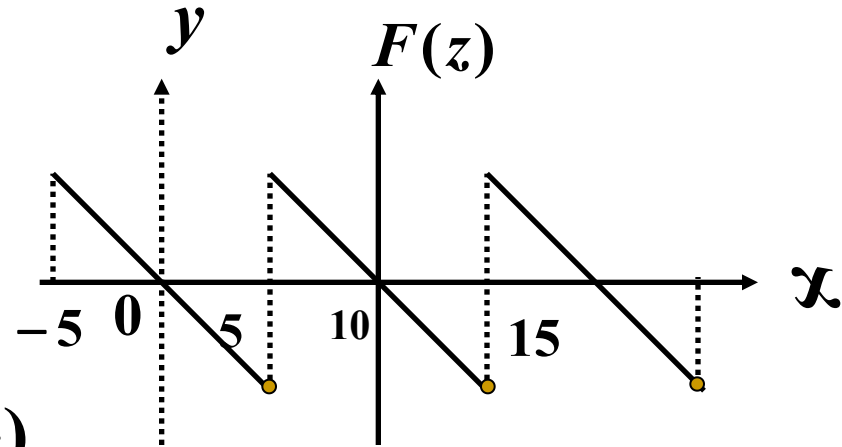
令 $F(-5) = 5$, 然后将 $F(z)$ 作周期延拓 ($T = 10$)

这拓广的周期函数满足 收敛定理的条件,

且展开式在 $(-5, 5)$ 内收敛于 $F(z)$.

$$a_n = 0, \quad (n = 0, 1, 2, \dots)$$

$$\begin{aligned} b_n &= \frac{2}{5} \int_0^5 (-z) \sin \frac{n\pi z}{5} dz \\ &= (-1)^n \frac{10}{n\pi}, \quad (n = 1, 2, \dots) \end{aligned}$$



$$F(z) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin \frac{n\pi z}{5}, \quad (-5 < z < 5)$$

$$\therefore 10 - x = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin \left[\frac{n\pi}{5} (x - 10) \right]$$

$$= \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin \frac{n\pi}{5} x. \quad (5 < x < 15)$$

另解 $a_n = \frac{1}{5} \int_5^{15} (10 - x) \cos \frac{n\pi x}{5} dx$

$$= 2 \int_5^{15} \cos \frac{n\pi x}{5} dx - \frac{1}{5} \int_5^{15} x \cos \frac{n\pi x}{5} dx = 0, (n = 1, 2, \dots)$$

$$a_0 = \frac{1}{5} \int_5^{15} (10 - x) dx = 0,$$

$$b_n = \frac{1}{5} \int_5^{15} (10 - x) \sin \frac{n\pi x}{5} dx = (-1)^n \frac{10}{n\pi}, (n = 1, 2, \dots)$$

$$\text{故 } f(x) = 10 - x = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin \frac{n\pi}{5} x$$

$$(5 < x < 15)$$